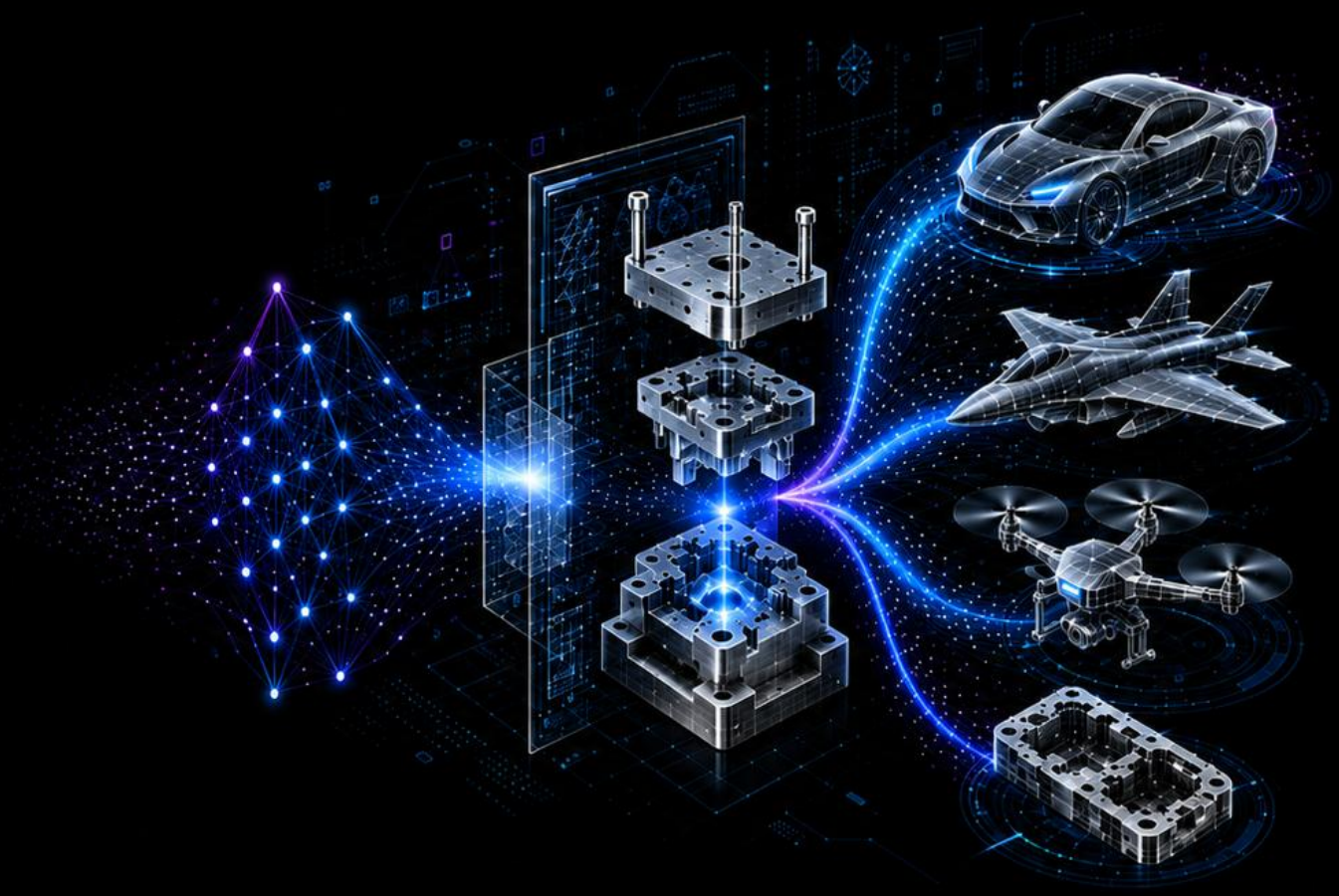
Study

Deep Learning



하형진

(지도교수: 양선웅)

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Intelligent Design for Engineering Applications Lab

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ANN & Deep Learning

- ANN에서 사용하는 구조는 퍼셉트론이라는 선형 분류기[오늘날 신경망(딥러닝)의 기원이 되는 알고리즘]
- 퍼셉트론은 다수의 신호를 입력을 받아 하나의 신호를 출력하는데 이 신호를 입력으로 받아 1 또는 0 정보를 앞으로 전달하는 원리

Logic Gate

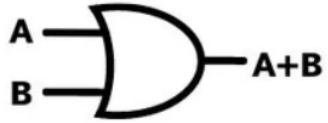
▼표 4-1 AND 게이트

x_1	x_2	y
0	0	0
1	0	0
0	1	0
1	1	1



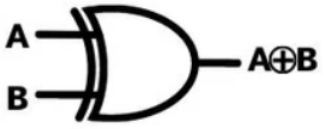
▼표 4-2 OR 게이트

x_1	x_2	y
0	0	0
1	0	1
0	1	1
1	1	1



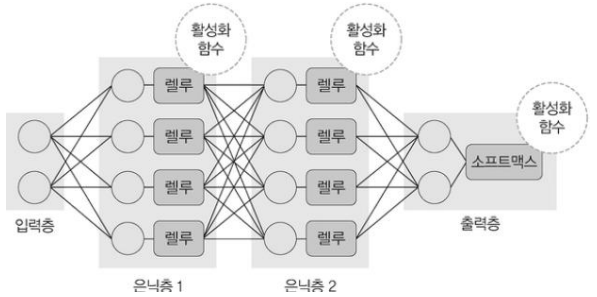
▼표 4-3 XOR 게이트

x_1	x_2	y
0	0	0
1	0	1
0	1	1
1	1	0



Deep Learing Term

❖ Deep Learning Structure



❖ Weight

❖ Bias

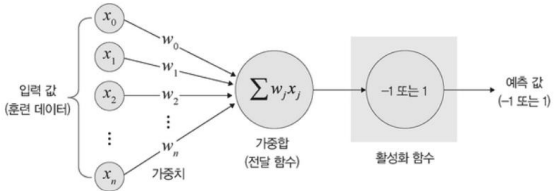
❖ Weighted Sum

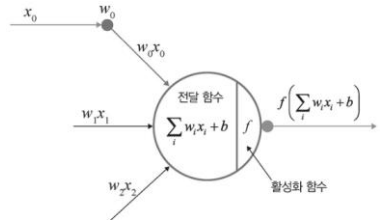
$$\sum_i w_i x_i + b$$

(w: 가중치, b: 바이어스)

❖ Activation Function

❖ Loss Function



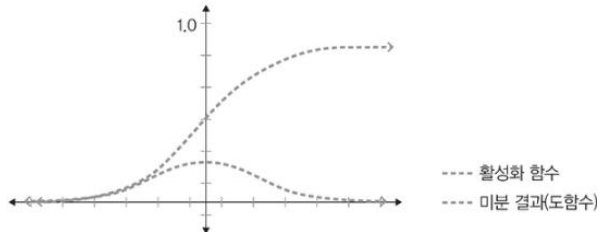


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Function

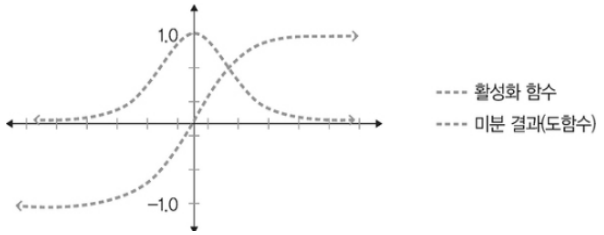
Activation Function

❖ Sigmoid

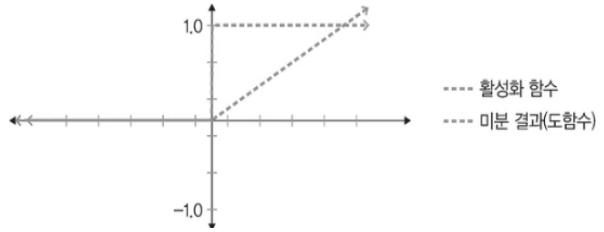


$$f(x) = \frac{1}{1+e^{-x}}$$

❖ Hyperbolic Tangent



❖ ReLU



Loss Function

❖ MSE(Mean Squared Error)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

($\hat{y}_i$: 신경망의 출력(신경망이 추정한 값)
y_i: 정답 레이블
i: 데이터의 차원 개수)

❖ CEE(Cross Entropy Error)

$$CrossEntropy = -\sum_{i=1}^{n} y_i \log \hat{y}_i$$

($\hat{y}_i$: 신경망의 출력(신경망이 추정한 값)
y_i: 정답 레이블
i: 데이터의 차원 개수)

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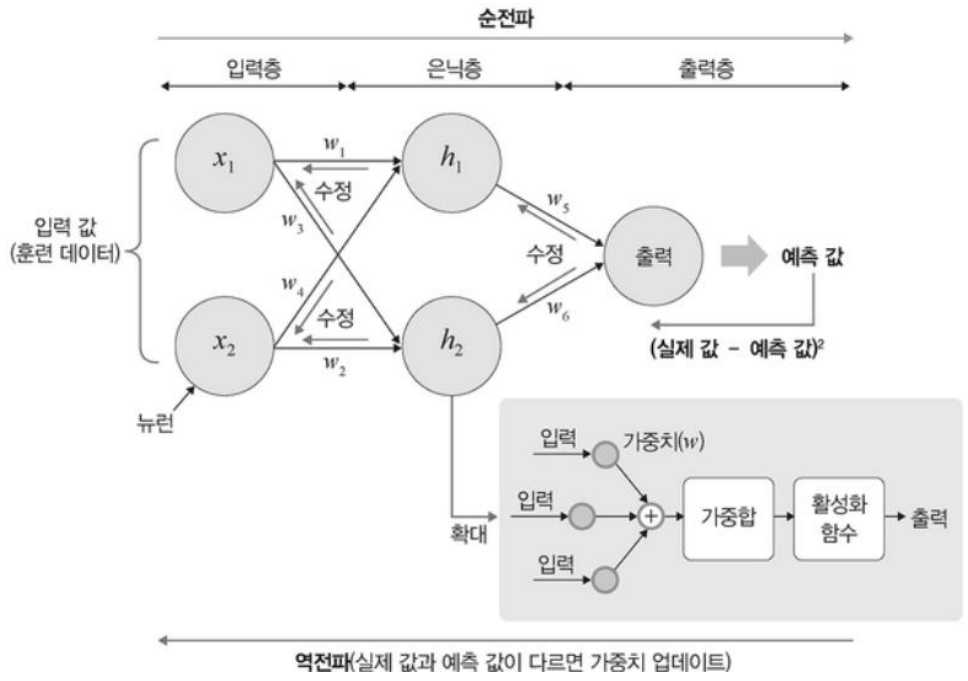
Deep Learning

- Deep Learning 학습은 크게 순전파와 역전파라는 두 단계로 진행
- Deep Learning 핵심과 문제 발생, 해결방안 탐구

FeedForward/BackPropagation

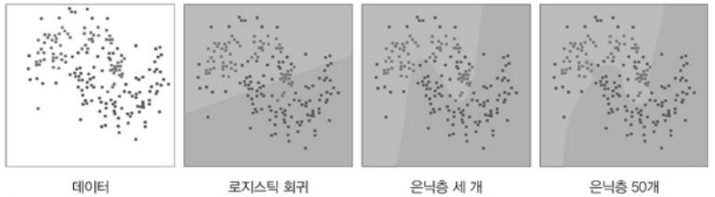
❖ FeedForward

❖ BackPropagation

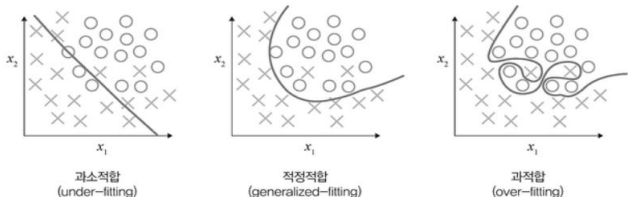


Core and Problems

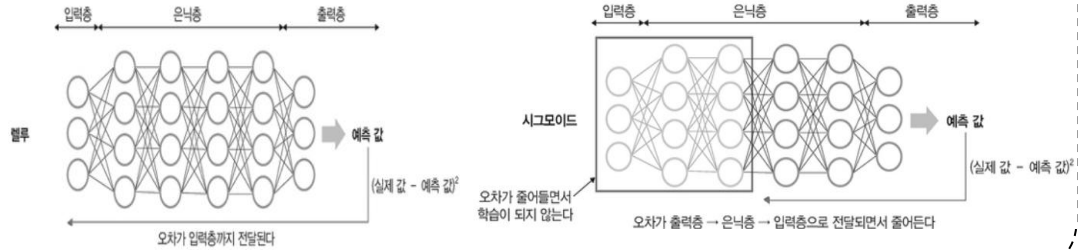
❖ Core



❖ Over-Fitting



❖ Vanishing Gradient



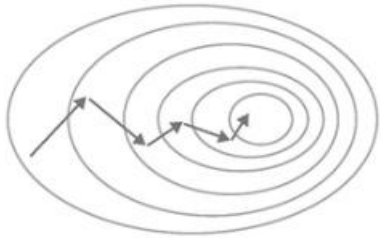
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Deep Learning

- Deep Learning의 문제점 해결 방안: Stochastic Gradient Descent, Mini-Batch Gradient Descent
- Deep Learning을 사용할 때 이점

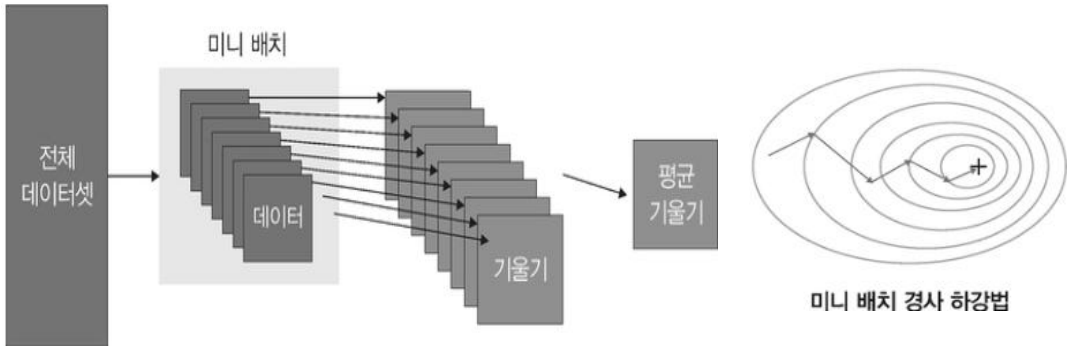
Solution

❖ Stochastic Gradient Descent



확률적 경사 하강법

❖ Mini-Batch Gradient Descent



미니 배치 경사 하강법

Advantage

❖ Feature Extraction

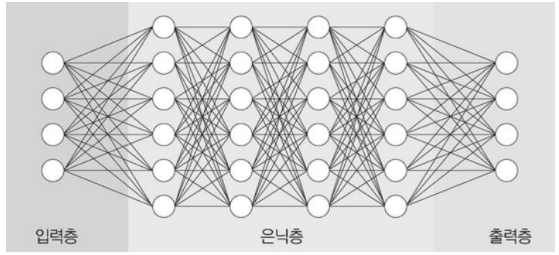
❖ Efficient use of big data

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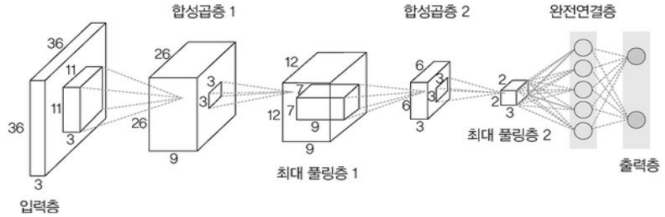
Deep Learning Algorithm

- Deep Learning Algorithm은 심층 신경망을 사용하는 공통점이 있음
- 목적에 따라 CNN, RNN, DBN 등으로 분류됨

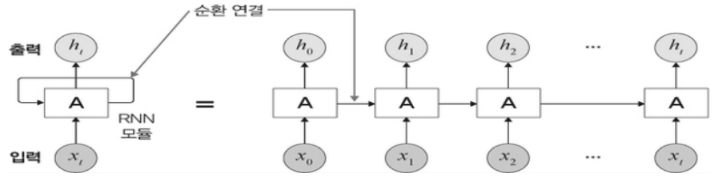
❖ DNN(Deep Neural Network)



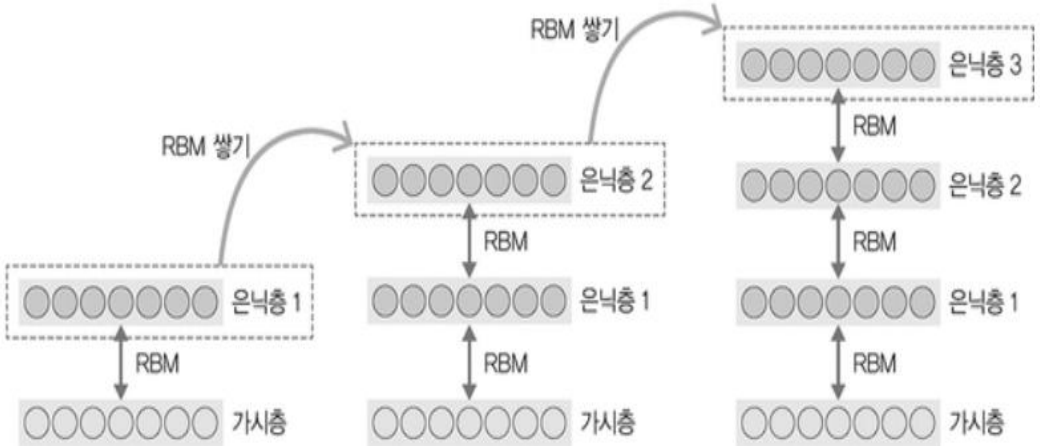
❖ CNN(Convolution Neural Network)



❖ RNN(Recurrent Neural Network)



❖ DBN(Deep Belief Network)



Any Questions?

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